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Return to sport and beyond following intramuscular tendon hamstring injury: A case report of an English Premier League football player

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ABSTRACT

Background: Hamstring strain injuries are the most common type of injury in elite football and are associated with a high risk of reinjury, particularly those involving the intramuscular tendon (IMT). Limited information is available regarding the rehabilitation and return to sport (RTS) processes following such injuries. This case study describes the clinical presentation of an elite football player following IMT hamstring injury, their on- and off-pitch rehabilitation alongside performance monitoring throughout RTS and beyond.

Case scenario: An elite football player suffered a grade 2c hamstring injury during an English Premier League (EPL) match. The player underwent early post-injury management, alongside progressive off-pitch physical preparation. The 'control-chaos continuum' was used as a framework for on-pitch rehabilitation to prepare the player for a return to full team training and competition. Objective and subjective markers of the player's response to progressive on- and off-pitch loading were monitored throughout RTS and beyond.

Outcomes: The player returned to on-pitch rehabilitation after 11 days, to full team training having achieved weekly pre-injury chronic running load outputs after 35 days and played in the EPL 40 days post-injury. The player did not suffer reinjury for the rest of the EPL season.

Conclusion: An understanding the unique structural and mechanical properties of the IMT, alongside expected RTS timeframes are important to inform rehabilitation and decision-making processes post-injury. Performance and frequent load-response monitoring throughout RTS and beyond, in conjunction with practitioner experience and effective communication are critical in facilitating effective RTS and reduce risk of reinjury following IMT injury.

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1. Introduction

Hamstring strain injuries (HSIs) are the most common type of soft tissue injury in elite football and may result in prolonged absence from competition [Ekstrand et al., 2016]. As well as being the most common injury in the sport, HSI's are also associated with a high risk of reinjury, particularly those involving the intramuscular tendon (IMT) [Shamji et al., 2021]. Despite increasing awareness of HSI's involving the IMT, their prevalence in elite football has not been reported [Brukner & Connell, 2016]. However, several authors suggest that the IMT is involved in approximately 20% of HSI's [Shamji et al., 2021; McAuley, Dobbin, Morgan, & Goodwin, 2022]. Damage to the IMT is regarded as more severe than a typical strain and has been associated with prolonged rehabilitation and return to sport (RTS) [Brukner et al., 2018; Pollock et al., 2016]. The IMT extends throughout the muscle belly, acting as a central supporting strut to which the muscle fibres

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attach [Pollock et al., 2014; Brukner & Connell, 2016]. Injury occurs in zipper-like fashion due to a failure of the myofibrillar attachments along the IMT, with oedema and blood products tracking alongside the torn muscle fibres [Brukner & Connell, 2016]. As a tendinous injury, the IMT has been classified as a type 'c' injury under the British Athletics Muscle Injury Classification (BAMIC) system. However, unlike the free tendon, retraction of the stumps of IMT lesions is limited since the surrounding muscle acts a splint, therefore surgical invention is rarely warranted [Brukner et al., 2018].

Despite current guidelines in the literature on RTS timelines following IMT hamstring injury in elite football and the high reinjury risk, limited information is available regarding the rehabilitation and RTS processes for such injuries [Shamji et al., 2021; McAuley et al., 2022]. We present the case of an English Premier League (EPL) and International football player who suffered an IMT hamstring injury during competition. We discuss the clinical presentation, magnetic resonance imaging (MRI) findings, and on- and off-pitch rehabilitation alongside performance monitoring throughout RTS and beyond.

1.1. Case scenario

The 23-year-old male football player, who gave informed consent for the use of his information, was a Bosnian International central midfielder with ~5 years playing experience across the Bundesliga (2010–2012), Nemzeti Bajnokság (2012–2014), EPL (2014–2016) and at International level since 2010. Nine minutes into an EPL fixture, the player felt a "sharp pain" after overstretching "but not excessively". The injury occurred whilst receiving a misplaced pass from one of his own central defenders, whilst being pressed by one of the opposing team's central midfielders inside his team's half of the pitch. Following the injury, the player was immediately removed from match-play. Post-match clinical examination by a team physiotherapist revealed a positive straight-leg raise (SLR) of 85^0 on the involved limb with altered sensation but no pain. Isolated hamstring contraction at $90^0/90^0$ produced pain but the player was pain-free during normal gait. The player underwent MRI ~48 h later to determine the full extent of the bleed and clearly define fluid collection with the muscle [Lee et al., 2012]. The MRI revealed a longitudinal partial-thickness split tear of the intramuscular portion of the conjoint tendon, demonstrating a long craniocaudal length tear involving the tendinous and aponeurotic portions. The associated myotendinous junction muscle belly injury involved both the biceps femoris and semitendinosus, classified as a BAMIC proximal to mid-grade 2c hamstring strain injury (Fig. 1).

1.2. Rehabilitation, return to sport and beyond

The player's RTS was viewed on a continuum, aligning his recovery and rehabilitation (Fig. 2) [Ardern et al., 2016]. The club's physiotherapy team delivered early care management using the POLICE (protection, 'optimal loading', ice, compression, elevation) principles [Bleakley et al., 2012]. After four days, the player regained full range of motion and was pain-free on both stretch and contraction. Based upon these clinical markers alongside positive feedback from the player, reconditioning began under sports science supervision with daily communication including pain response (numerical rating scale; NRS) and periodic checks of the player with a member of the club's physiotherapy team. 'Optimal loading' continued (Glasgow et al., 2015), transitioning from submaximal isometrics (under initial physiotherapy care) to maximal isometrics (overcoming, with the later addition of yielding derivatives) as the predominant mode of strength training during initial programming [Taberner, Haddad, et al., 2020; Taberner & Cohen, 2018]. Six days post-injury the player performed a unilateral isometric posterior chain (IPC) test (Fig. 3) to quantify the players' ability and willingness to produce maximal force (IPC peak force', IPC-PF; Newtons) [Taberner, Haddad, et al., 2020; Taberner & Cohen, 2018]. This strength diagnostic tool was used from this point onwards to quantify weekly progress throughout rehabilitation and more frequently after RTS to assess load response via absolute outputs in the involved limb and changes relative to the non-involved limb, alongside % interlimb asymmetry (%ILA). The latter data, characterising the "residual response (to on- and off-pitch loading)" reflecting mechanical loading patterns within a session or series of sessions was used to inform progression decisions

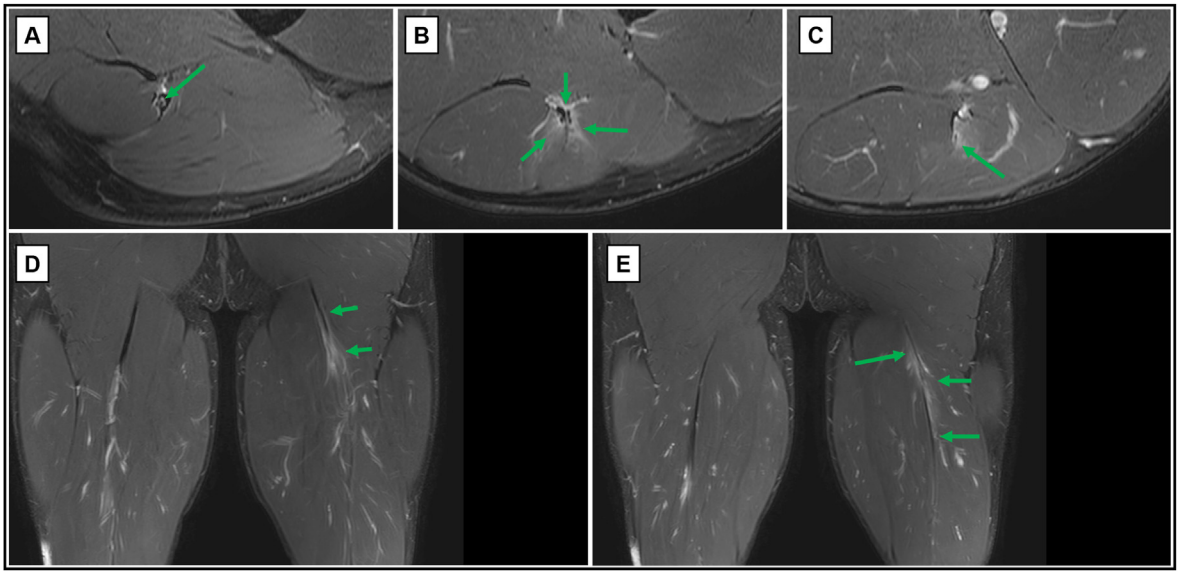

Fig. 1. Post-injury magnetic resonance imaging (MRI) classified as a BAMIC proximal and mid-grade 2c hamstring injury. Green arrows indicate key prognostic features. (A) Intrasubstance longitudinal split tear of conjoint tendon. (B) Conjoint tendon split and MTP junction injury of biceps femoris and semitendinosus. (C) Split tear extension into aponeurotic portion of the intramuscular tendon. (D) Craniocaudal extent of the injury. (E) Craniocaudal extent of MT junction haemorrhage. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

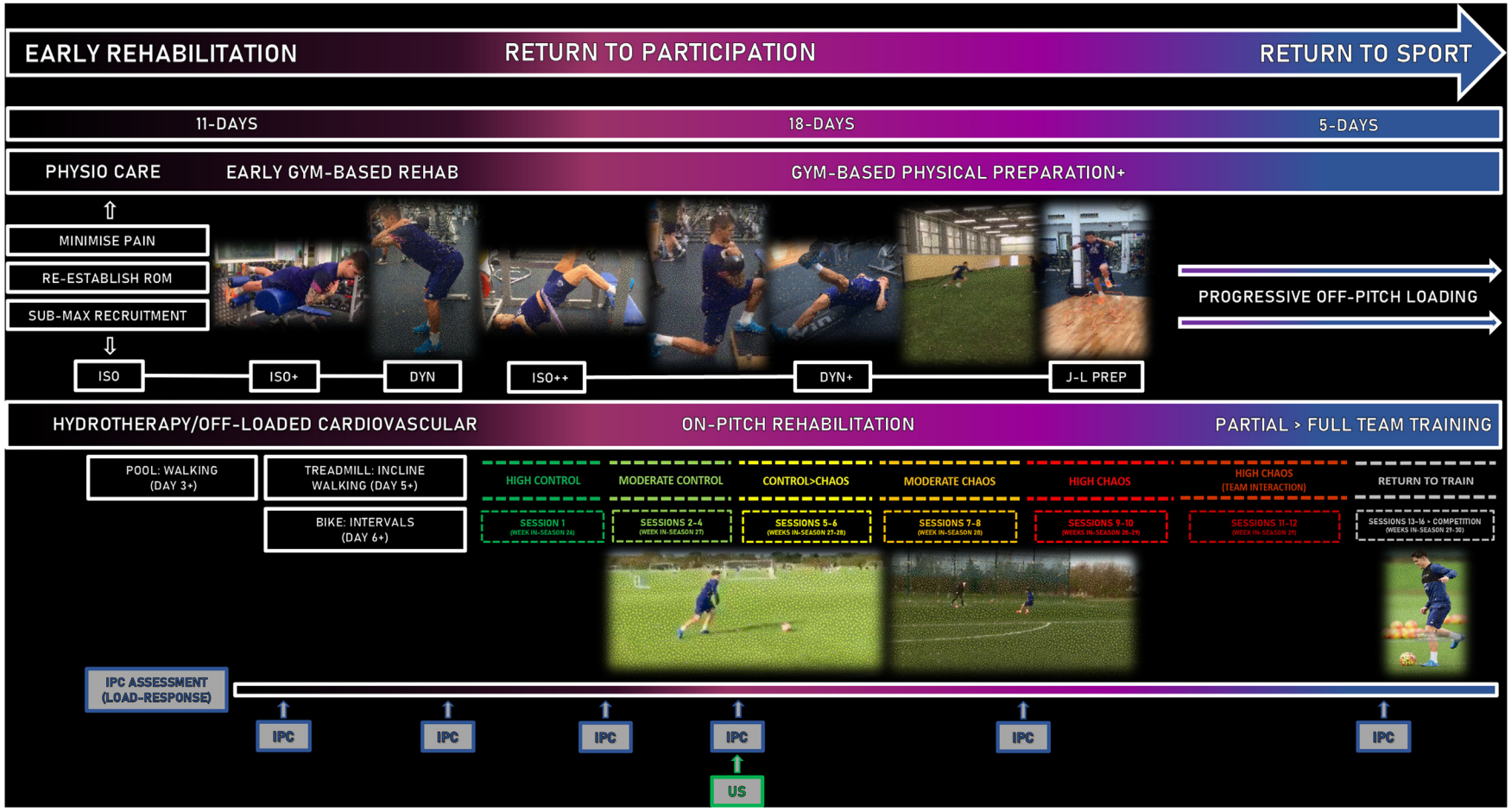

Fig. 2. Overview of the Return to Sport (RTS) of an elite male football player following a BAMIC proximal and mid-grade 2c hamstring injury framed upon the a modified RTS continuum [Ardern et al., 2016]. Early rehabilitation (actively involved in rehabilitation but not at level for on-pitch rehabilitation) divided into two sub-phases: physiotherapy care (physio care) and early gym-based rehabilitation (11-days) including hydrotherapy (pool – walking)/off-loaded cardiovascular conditioning (incline walking and bike aerobic interval training) and sub-maximal (ISO) > maximal isometrics (overcoming to yielding) (ISO+) and early dynamic strength training (DYN)(bodyweight and low-load resistance). Return to participation phase (involved in on-pitch rehabilitation but not a level to return to sport) (18-days); on-pitch sports-specific reconditioning using the 'control-chaos continuum' (CCC) (number of sessions within each phase displayed and respective in-season week number) plus progression of gym-based physical preparation; ISO++, DYN+ and jump-landing preparation (J-L PREP). Return to participation amalgamates with partial RTS (at a level to return to partial team training) by bridging a return to full team training (Return to Train) via adapting the last phase of the CCC; High chaos – team interaction [Taberner, Allen, et al., 2020]. Arrows indicate timepoints of load response monitoring using a unilateral isometric posterior chain (IPC) test on a portable force platform sampling at 1000 Hz (PS-2141, Pasco, Roseville, CA) with proprietary software (ForceDecks, NMP technologies, London, England) and ultrasound (US) assessment.

Fig. 3. Isometric Posterior Chain (IPC) test performed using a portable force platform sampling at 1000 Hz (PS-2141, Pasco, Roseville, CA) with proprietary software (ForceDecks, NMP technologies, London, England). Test performed with the player lying supine position, the heel of the testing limb placed on the force platform resting on a firm plinth with the testing angle set at 90° hip and knee flexion using a goniometer (Physio Parts, Twickenham, UK) and the non-testing limb relaxed and fully extended. The player was instructed to push the heel of the testing limb into the force platform exerting as much force as fast as possible whilst keeping the buttocks, hips and head on the mat and hands crossed on chest. External pressure was applied to the non-tested pelvis/hip to prevent hip extension. The player was familiar with the test, but standardised instructions were given before the assessment, a verbal countdown of "3, 2, 1" before the initiation of a maximal effort contraction which was held for 3-s during which consistent verbal encouragement was given "PUSH, PUSH, PUSH, RELAX". 3x maximal voluntary contractions were performed with 10s rest between repetitions.

alongside achievement of running load targets (using global positioning systems (GPS)) on the player's return to on-pitch reconditioning [Taberner, Haddad, et al., 2020; Taberner et al., 2019; Taberner & Cohen, 2018]. These objective measures were used in conjunction with absence of pain (NRS) and other feedback from the player, as key components in the decision-making process [Taberner, Haddad, et al., 2020; Taberner et al., 2021]. Following six days of periodised programming to allow adaptation (adequate rest windows between activity), absolute IPC-PF output on the involved limb had improved substantially, likely due to the player's increased confidence and willingness to apply force. As a consequence peak force %ILA was now <10% (Table 1).

Having achieved this criterion alongside an absence of the progressive tightness or acute pain associated with IMT injury [Brukner et al., 2018], the player returned to on-pitch running the following day (Figs. 2 and 4). The 'control-chaos continuum' (CCC) was used as a framework for progressive on-pitch training prescription, adapted to the injury-type, the specific requirements of the player's position and inter-related contextual factors [Taberner et al., 2019; Taberner, Allen, et al., 2022]. Following four on-pitch training sessions (high and moderate control) and progressive off-pitch loading consisting of isometrics complemented with dynamic strength training, using an increased volume load on the involved limb where appropriate (i.e., 3-4 sets involved limb/2 sets non-involved limb [Taberner, Haddad, et al., 2020; Taberner & Cohen, 2018], the player's structural response to this early loading and healing was assessed with ultrasound (US) (Fig. 2; 19 days post-injury). The radiologist reported resolution of peritendinous oedema around the conjoint tendon and mature scar tissue formation in the region of longitudinal splitting. Furthermore, the myotendinous junction muscle belly showed signs of remodelling, aligning with the expected appearance at this stage post injury. No complicating features were noted. At this stage of healing, involved limb IPC-PF output had improved, with % ILA <10% (Week 27; Fig. 2 and Table 1).

Nineteen days post injury, the player transitioned into football-specific phases of the CCC (control to chaos phase onwards), with the objectives of progressive training reflective of the content and structure of the team environment alongside systematic exposure to higher speeds and high-speed running volumes - representative of his gameload (mean output) [Taberner et al., 2019]. Training content such pattern of play drills were designed specifically to reflect the player's positional movement demands, including

Table 1
Unilateral Isometric Posterior Chain (IPC) test data performed on a portable force platform sampling at 1000 Hz (PS-2141, Pasco, Roseville, CA) with proprietary software (ForceDecks, NMP technologies, London, England). Descriptives include mean peak force (PF), standard deviation (SD) and inter-trial percentage coefficient of variation (%CV). O–P = Off-pitch Rehabilitation, Pre R1 = On-Pitch Rehabilitation Session 1, R4 = On-pitch Rehabilitation Session R4, MD-1 = Match-day minus 1, MD-2 = Match-day minus 2, MD-4 = Match-day minus 4, MD-5 = Match-day minus 5, (2) second session type within same training week. Coding links to Fig. 3.

In-season Week Number	Day of Week	Session Type	Mean IPC-PF (Right)	SD IPC-PF (Right)	% CV (Right)	Mean IPC-PF (Left)	SD IPC-PF (Left)	% CV (Left)	Mean IPC-PF % ILA
25	Sat	O–P	215	1	1	106	11	11	51
26	Thu	O–P	205	2	1	185	2	1	10
27	Mon	Pre R1	236	14	6	263	8	3	−11
28	Fri	Pre R4	233	18	8	244	4	2	−5
30	Mon	Pre MD-1	236	6	2	245	14	6	−4
30	Thu	Pre MD-2	265	16	6	270	7	3	−2
30	Fri	Pre MD-1 (2)	255	7	3	256	2	1	0
31	Mon	Pre MD-5	246	5	2	247	2	1	0
31	Tue	Pre MD-4	216	1	0	199	8	4	8
31	Thu	Pre MD-2	253	4	2	277	13	5	−9
31	Fri	Pre MD-1	232	3	1	228	8	4	2
32	Mon	Pre MD-5	246	4	2	238	8	4	3
32	Tue	Pre MD-4	222	19	9	217	6	3	2
32	Thu	Pre MD-2	256	3	1	225	1	1	12
32	Fri	Pre MD-1	252	17	7	233	6	2	8

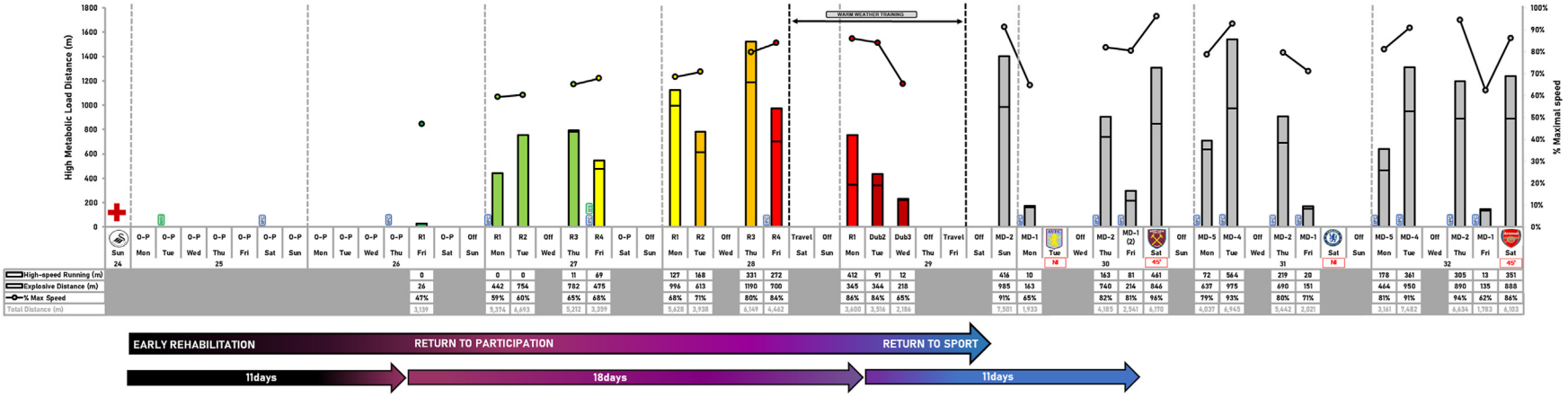

Fig. 4. High-speed running (HSR), explosive distance (Exp-D), maximal speed exposure and total distance covered during the return to sport and return to team training/competition after intramuscular hamstring tendon injury. Session HSR (top segment of stacked bar)(>5.5ms^{-1}; dwell time 0.5s) distance, Exp-D (bottom segment of stacked bar)(distance covered accelerating/decelerating from 2 to 4 ms^{-1} <1 s) and combined equals high metabolic load distance (distance above 25 w•kg^{-1}; HSR plus Exp-D) represented on the left y-axis with exact values represented in the table below including total distance (m) covered per session. The % max speed (right y-axis) is % of pre-injury maximal speed achieved within session. Data collected in training using global positioning systems (10 Hz Viper, StatSports, Belfast, UK) and match data collected using TRACAB Gen4 optical tracking sampling at 25 Hz (ChyronHego, New York, USA) – interchangeability between technologies as determined by Taberner, O'Keefe, et al., (2020). Colour-coded bars represent phases of the 'control-chaos continuum' (Taberner et al., 2019; Taberner, Allen, et al., 2020); dark green = high control, light green = moderate control, yellow = control > chaos, orange = moderate chaos, red = high chaos, dark red = high chaos (team interaction) and grey = return to full team training. Bar underneath graph represents modified return to sport continnum (Ardern et al., 2016); black = early rehabilitation – player in off-pitch rehabilitation and not actively at the stage to return to on-pitch activities, purple (graded) = return to participation – player in rehabilitation actively participating in both on- and off-pitch rehabilitation but not ready to return to full team training, blue (graded) = return to sport – player actively involved in team training (partial to full involvement). Club badges indicate opposition faced in competition (Aston Villa = FA Cup, Swansea, West Ham United, Chelsea and Arsenal = English Premier League) and red labels indicate number of mins player or not-involved (NI = unused substitute). Grey dashed bars dictate week to week loading frames, black dashed bars indicate warm weather training camp in Dubai. Green logos = imaging time points (MRI and US) Blue logo = unilateral isometric posterior chain (IPC) test on a portable force platform sampling at 1000 Hz (PS-2141, Pasco, Roseville, CA) with proprietary software (ForceDecks, NMP technologies, London, England) performed prior to training each morning. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

technical skills and energy system conditioning [Taberner et al., 2019]. An IPC assessment prior to the player joining the team on warm weather training camp in Dubai (in-season week 28–29) and transitioning into the high chaos phase, showed stable involved limb IPC-PF and %ILA (Table 1), suggesting no adverse response to the preceding on- and off-pitch activities. Following travel and a planned day-off for acclimatisation to the new environment, the player performed his last rehabilitation session away from the team, with the goal of exposing the player to >70% gameload HSR volume under chaotic conditions (extensive positional conditioning) and challenging the player to receive passes in a similar manner as the injury mechanism; during pass and move drills designed to replicate this demand on the player [Taberner et al., 2019; Allen et al., 2021]. For the next two rehabilitation sessions (30 days post-injury) the player rejoined partial team interaction (high chaos: team interaction; Figs. 2 and 4) [Taberner, Allen, et al., 2020] as following discussion with the coaching staff, the session content was deemed appropriate for the player to return to training with the planned volume, intensity and duration part of a planned team deload after a congested festive fixture period in the EPL [Taberner et al., 2019; Taberner, Allen, et al., 2022].

After the warm weather break (35 days post-injury), the player had achieved preinjury weekly chronic running load volume targets (total distance and HSR), achieved above 70% relative HSR volume within a single rehabilitation session and >85% maximal speed, and had been challenged in relation to the demands of team training and the mechanism of injury [Taberner et al., 2019]. The player reported no awareness of pain or loss of tension and critically felt confident about returning to full team training [Brukner et al., 2018; Taberner et al., 2021]. The performance and medical team were cognisant of the evidence showing high reinjury risk following this particular grade of injury [Pollock et al., 2016; Shamji et al., 2021]. However, communication with the coaching staff indicated that the player was unlikely to be selected immediately for match-play. As such, the shared decision-making team allowed the player to return to full team training, deeming that this would expose the player to a reinjury risk that was below their risk tolerance assessment [Taberner et al., 2019; Taberner, Allen, et al., 2020; Taberner, Cohen, et al., 2022]. The match-day minus two training session (MD-2; week 30) provided a greater training stimulus than previous training sessions from a running load and ecological perspective (technical, tactical, decision-making and associated perceptual challenges), and included extensive tactical games in preparation for team's upcoming fixture in the FA Cup. As part of the performance monitoring process post RTS, particularly with this injury type, the player completed an IPC test each morning prior to team training, with its purpose explained to the player who was comfortable with performing these assessments [Taberner et al., 2021]. Compared to his previous test, the match-day minus-one (MD-1) IPC assessment – reflecting the response to the MD-2 session – showed stable absolute outputs and %ILA– suggesting no adverse response on the involved limb. Following the MD-1 session, a player due to be involved in the match-day squad became ill, resulting in the returning player being added to the match-day squad as cover, but was ultimately an unused substitute. In a subsequent pre MD-2 session IPC assessment which followed a scheduled team day off, both involved and non-involved limb PF increased, potentially indicative of 'supercompensation' (four days post MD-2; week 30) – following the most demanding session since the start of on-pitch training post-injury for the player, while ILA was now 5%. The MD-2 training session running load outputs were within 'normal' boundaries (+/– 1SD) for the player and no negative responses were identified in subsequent MD-1(2) IPC testing (Fig. 2 and Table 1), and the player reported no symptoms of acute tightness, pain, or loss of tension. Following a typical low volume MD-1 training session with tactical emphasis, the player was selected for the match-day squad as a substitute. However, as a result of a red card in the first half, the manager made a tactical decision at half-time to bring on the player to provide extra cover for the defence, typically referred to in football terminology as a 'layer of protection' in front of a back three. Subsequently, as the team were down to ten men, the player's running load output was substantially above expected gameload for ~45mins match-play (~0.80 gameload HSR; maximal speed; 8.75ms^{-1}, 96% of the players maximal speed). Post-game, the player reported no symptoms. Typically, the ~24-48hr period post-game has been associated with altered isometric posterior chain performance [Nedelec et al., 2014; Constantine et al., 2019], therefore the decrement in IPC-PF on both limbs (while remaining <10% ILA) pre MD-2 (week 31) represented an expected response in the context of his running load outputs in the game (HSR per/min). Over the subsequent two-week period (Fig. 4; in-season weeks 31–32), the player's training load running outputs (within MD-type and weekly chronic load) were monitored, ensuring that training included sufficient exposure to HSR volume and maximal speed efforts. Pre-training, IPC testing was routinely performed to identify any concerning alterations in force production capacity, alongside daily communication with the player, with no negative symptoms reported. The player also continued with focal loading (overcoming and yielding isometrics alongside dynamic strength) with the specific content and volume load dependent upon the structure and training load demands of the on-pitch training week. The player did not suffer reinjury during the remainder of the season, and mainly featured as a substitute (5x games; ~40–45mins) apart from one start in an EPL game, completing ~90 min (TD; 11,430m, HSR; 719m, 8.36 ms^{-1}, 92% maximal speed). This HSR output (1.26x gameload) was above his preinjury typical HSR gameload, a very positive sign in the context that HSR outputs are reported to be suppressed after HSI in football [Whiteley et al., 2021]. However, match running load outputs should always be intepreted alongside contextual factors related to the players role in the team, tactics, opposition, and the score-line [Paul et al., 2015].

2. Discussion

2.1. Return after intramusculur tendon injury and reinjury risk

In track and field, HSI's extending into the tendon, classified using the BAMIC system as type 'c' injuries, were observed to be both more prone to reinjury and took athletes longer to RTS [Pollock et al., 2016]. In spite of this, the expected timeframe for RTS after IMT injury and reinjury risk remains inconclusive. Furthermore, the relevance of IMT injuries has been questioned, with complete resolution of IMT injuries on MRI not required for RTS [van der Made, Almusa, Reurink, & et al., 2018; van der Made, Almusa, Reurink, & et al., 2018]. However, it has been proposed that sequential MRI may be a useful tool to inform on healing status and thus help guide rehabilitation [Baldock et al., 2021].

In a recent longitudinal study over 8 seasons in an EPL club, HSI's involving the IMT (13 out of 61) resulted in a longer time to return to full training (mean 36 days versus 24 days without involvement) and a higher reinjury rate (38.5% versus 12.5%)[Shamji et al., 2021]. However, in another EPL club, across 4 seasons no differences in return to training times or rates of reinjury upon return were found between HSI's with (6/40 HSI's) or without IMT involvement [McAuley et al., 2022]. Interestingly, in both studies, type 2c HSI's were associated with the longest return to full training across all the grades examined. In the present case report, the player returned to partial team training after 29 days, full team training after 35 days, played 45 min as a substitute after 40 days and did

not suffer a reinjury upon return to training or competition. This timeframe of return to full training aligns with that reported by Shamji et al., [2021] following IMT injury, and also within the 95% percentile of those reported by McAuley et al. [2022]. However, due to the large variability in specific injury presentation between players, healing timeframes, rehabilitation processes and neuromuscular response to loading, each injury should be treated on an individual basis with the available evidence only providing a guide for expected RTS.

2.2. The intramuscular tendon and 'optimal loading'

While similarities exist between the collagenous make-up of the free tendon and IMT, structural and mechanical differences should influence the 'optimal loading' approach during the rehabilitation process [Brukner et al., 2018]. The free tendon can tolerate high strain rates which are stored and released during stretch-shortening activity, with movement mostly derived between the fascicles within the intermuscular matrix rather than the collagen fibres [Screen, Bader, Lee, & Shelton, 2014]. In contrast, the IMT which is primarily comprised of type I collagen is significantly stiffer due to a lack of fascicular bundles and interfascicular matrix, and thus cannot store and release energy like the free tendon. IMT injury involves haemorrhage, subsequent inflammatory responses and proliferation resulting in hypertrophic IMT scar issue. This scar tissue is 'functionally' immature during early rehabilitation but as remodelling and scar tissue maturation occurs throughout rehabilitation and beyond, type III collagen and extracellular matrix are replaced with longitudinally orientated type I collagen, restoring tensile strength and tendon stiffness [Brukner et al., 2018; James et al., 2008; Liu et al., 1995]. The use of isometric loading, transitioning progressively from sub-maximal to maximal overcoming derivatives (adjusted according to pain response both during and post-session alongside willingness to apply force), complemented with high strain yielding variations potentially develop tendon stiffness, and are therefore warranted following IMT injury. [Schaefer & Bittmann, 2017; Taberner, Haddad, et al., 2020; Taberner & Cohen, 2018]. From a loading prescription perspective, intensities >80% of maximal voluntary contraction are required to stimulate the development of mechanical and material tendon properties. We programmed utilising clusters sets (3–5 × ~3–5 s iso-holds) to ensure that absolute intensity was maintained within sets, with the conceptual goals of developing maximal strength, RFD and strength-endurance qualities (please see Taberner and Cohen (2018) and Taberner, Haddad, et al. [2020] for video content regarding specific exercise selection) [Bohm et al., 2015; Bohm et al., 2019; MacDonald et al., 2018]. Conceptually, the use of yielding isometrics – holding a fixed position and resisting deformity to further eccentric lengthening offers an important loading medium following IMT injury. High strain oscillations generated by the contracting muscle as a result of resisting motion are transmitted to the tendon, with the tendon acting as passive force transmitter [Schaefer & Bittmann, 2017; Schaefer & Bittmann, 2018]. Theoretically, this mechanical strain causes longitudinal deformation of the tendon as a result of contraction by the adjacent muscle [Bohm et al., 2019]. The external strain is transmitted through the extracellular matrix to the cytoskeleton of the mechanosensitive tendocytes, initiating expression of genes and growth factors responsible for catabolic and/or anabolic cellular and molecular responses such as collagen synthesis [Wang, 2006].

Dynamic strength training was also progressively integrated into the player's off-pitch rehabilitation, beginning with knee > hip dominant exercises before adding more hip dominant exercises to target the proximal injury site. Eccentric exercise derivatives i.e., RDL, sliding leg curl and its variations were progressively integrated within programing to ensure sufficient mechanical strain whilst ensuring volume and intensity were not 'excessive' during early rehabilitation since it may it may impair tendon healing [Taberner et al., 2016; MacDonald et al., 2019]. Dynamic exercises were also progressively focused upon targeting intermuscular adaptations and the proximal to distal activation pattern observed during the stance phase of gait, optimising energy transfer from hip to knee and from knee to ankle, with the conceptual goal of increasing transfer to the coordination demands of HSR [Taberner & Cohen, 2018]. Furthermore, ample rest-recovery windows were periodised between off-pitch loading sessions (~36–72h apart) to ensure muscle-tendon adaptation [Macdonald et al., 2019], a critical factor in a player's recovery and rehabilitation following this type of injury.

2.3. Performance monitoring and on-pitch rehabilitation

Following injury, the player's reported pain on clinical examination (particularly contraction) resolved relatively quickly. This observation aligns with evidence indicating that if the IMT injury is partial, with parts of the contractile element of the muscle–tendon unit remaining intact, force production in low-level tasks or clinical assessments may return quickly [MacDonald et al., 2019]. After a short period of early isometrics (submaximal – overcoming) under the supervision of the physiotherapy team, the player was assessed with a supine 90^0 unilateral IPC test – the use of which we have previously described in the rehabilitation and RTS of elite football players after HSI [Taberner, Haddad, et al., 2020; Taberner & Cohen, 2018]. This strength diagnostic tool has been highlighted as a reliable tool to assess posterior chain force production and sensitive to fatigue induced by football match-play [McCall et al., 2015; Constantine et al., 2019]. Initial deficits in involved limb force production, relative to the contralateral limb were not unexpected (Table 1). In addition to capacity losses associated with the mechanical effects of the injury, willingness to exert maximal effort can be compromised in the early stages of rehabilitation, possibly due to apprehension or fear of making the injury worse, alongside structural damage [Kvist & Silbernagel, 2022]. Therefore, the gradual improvements in IPC-PF outputs were considered to be due to a combination of enhanced capacity, and the player's greater willingness and confidence in applying maximal force through the involved limb, driving a reduction of %ILA to <10% by the eleventh day post-injury (Table 1). We have previously used this criterion while also considering progression of involved limb absolute outputs alongside the absence of pain, a range of motion within normal limits in both limbs, and player feedback to inform the decision-making process to return to running under conditions of high control[Taberner & Cohen, 2018; Taberner, Haddad, et al., 2020; Taberner et al., 2019]. Absence of pain, range of motion, and strength appear to key measures to support return to running after HSI [van der Horst et al., 2017; Dunlop et al., 2020]. While the optimal measurement of strength (isometric, isotonic (concentric, eccentric) isokinetic) still remains to be determined [Dunlop et al., 2020], evidence supports the use of IPC-PF as an indicator of neuromuscular status in football players [Taberner & Cohen, 2018; Taberner et al., 2018; Matinlauri et al., 2019]. More specifically, it has been used to quantify ILA, assess chronic changes in hamstring strength and response to off- and on-pitch loading in elite players during the rehabilitation of hamstring injuries [Cuthbert et al., 2021; Taberner & Cohen, 2018; Taberner, Haddad, et al., 2020]. Practical factors such as the low time cost of the assessment, the limited risk of muscle damage or soreness associated with this overcoming type of isometric contraction – with possible potentiation effects on subsequent activity, also facilitate its frequent use in this context.

The specifics of an injury case are used to determine the number of sessions within each phase of the CCC [Taberner et al., 2019; Taberner, Allen, et al., 2022]. In the present case, we transitioned the player from high to moderate control rapidly, introducing directional change load and low volume-intensity technical skill integration alongside linear running (<65% maximal speed) [Taberner et al., 2019; Allen et al., 2021; Taberner, Allen, et al., 2022]. Additional imaging (US), and an IPC-test showing no negative features or responses were key criteria clearing the player's transition into the football-specific phases of the CCC (control to chaos phase onwards). In these phases, HSR running volumes were progressed (0.12, 0.22, and 0.30 gameload initially) before exposure to higher volumes (0.47, 0.58 and 0.72 gameload) prior to return to partial team training. Furthermore, HSR volume targets was gradually achieved moving from controlled aerobic-power interval running into positional-specific context such as recovery runs following the break-down of an attacking transition. On-pitch rehabilitation, however, did not only focus upon achieving target running outputs, session content was also structured to align with team training content i.e., warm-up, pass and move etc, thereby integrating the ecological demands the player would face upon RTS [Taberner, Allen, et al., 2022; Taberner et al., 2019]. In addition, drill design incorporated receiving passes in the same manner of the injury mechanism, intrinsically developing the player's self-confidence without directly discussing this with the player[Allen et al., 2021]. Partial team training during the warm weather training camp also allowed the player to rejoin the group and be exposed to football-specific 'chaos', albeit with the player's level of involvement discussed and agreed with the coaching team. Prior to RTS, the player had achieved pre-injury mean weekly chronic running load targets, preparing him for a return to full team training. This was considered an appropriate target since he was not typically in the starting line-up, therefore the shared decision-making team did not view targetting concurrent load outputs inclusive of match-play as appropriate targets for this particular case [Taberner, Allen, et al., 2022]. Monitoring and managing training load has been previously highlighted as an important factor in practitioners RTS decision-making processes as it is representative of an athlete's training capacity in relation to the demand of the sport [Dunlop et al., 2020; Taberner, Cohen, et al., 2022].

Given the reinjury rates observed following IMT injury (Pollock et al., 2016; Shamji et al., 2021), the continuing use of the IPC test as a tool to assess acute and residual response to load in the early post-RTS period was considered another important part of the decision-making process. Performance monitoring post-RTS and implementation prior to each team training session to objectively quantify the neuromuscular response to the previous days loading is a novel concept, not reported in the literature. As strength recovery is recognised as one of the most important factors in both return to running and RTS decision-making[Dunlop et al., 2020], a greater emphasis on neuromuscular monitoring is logical during this period of increased vulnerability [Stares et al., 2019] as a means to obtain insights on response beyond that of subjective feedback from the player. The player's running load outputs were also closely monitored post-RTS, ensuring HSR volume was at least maintained from load accumulated in football-specific context typically coming from the MD-4 training session under the coaching team's training methodology and/or competition. The largest decline in total IPC-PF output was observed at ~48hrs following the player's ~45 min as a substitute in the EPL game against West Ham United (Table 1 and Fig. 1), broadly aligning with the timing of peak neuromuscular fatigue responses to match-play in elite players ~48-72hrs post-match [Nedelec et al., 2014]. Nonetheless, despite the decline in summed total force output across both limbs, this trend was symmetrical. We interpret the symmetrical fatigue (parallel trends on involved and non-involved limb) to reflect a willingness to load and therefore generate fatigue in both limbs during on-pitch activities, such as during HSR – with large musculotendon strain upon the posterior chain [Schache et al., 2012]. Conceptually, this underpins the interest in frequent monitoring of individual limb responses to recent loading demands via consideration of changes in absolute outputs alongside %ILA under conditions of acute and/or residual fatigue, in addition to periodic assessments of "chronic adaptations" to reconditioning – ideally assessed in 'recovered' conditions. As such, symmetrical fatigue can be considered a positive sign, aligning with the player reporting no unusual discomfort during or in the subsequent day(s) following the activity. Hence, it was deemed not necessary to amend his involvement in training in the forthcoming week.

This case study provides an insight into the multiple factors, objective and subjective data which inform the shared decision-making process during the RTS of an elite footballer following IMT hamstring injury. However, the acute injury presentation, on- and off-pitch rehabilitation loading progressions, response to load, healing and RTS timeframes are player-specific and may not directly transfer to other elite footballers. Note for example, that in the present case, the coaching team allowed the player to return to partial team integration to further accustom the player to team training demands following injury, which may not always be feasible [Taberner, Allen, et al., 2020]. This will depend upon the club or level of competition (i.e., academy or First Team), phase of the season, or the coaching team's approach – they may not allow a player to return to team training until the risk tolerance is deemed to be at an acceptable level for them to train without restriction. Nonetheless, the concepts (i.e., principles and processes) discussed with this case report are an adaptable frame that the practitioner would modify to the requirements of the respective player, their injury case, clinical presentation alongside the dynamics of the club. The frequent subjective and objective monitoring of player status enhances the characterisation of responses, thereby informing case-specific adjustments.

3. Conclusion

Hamstring strain injuries are the most prevalent type of injury in elite football, with a high risk of reinjury especially if the IMT is involved. Despite similarities with the free tendon, the IMT has unique structural and mechanical differences which warrant special consideration to ensure 'optimal loading'. Awareness of these differences may also influence decision-making on progression of on- and off-pitch loading, further informed by objective data, clinical experience, and player feedback. Furthermore, following IMT injury off-pitch conditioning, performance and load-response monitoring should continue beyond RTS, supporting efforts to reduce the elevated risk of reinjury.

List of co-authors and their contribution

MT wrote the initial draft. MT and DDC edited and adapted the initial draft. MT, DDC, and JOK continued editing until a final version of the manuscript was agreed. AD checked radiography details within the manuscript.

Consent

Informed consent provided by player and parent club (club headed letters)

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Ethical statement

No ethical approval was required for this case study. Written consent provided by club and player.

Declaration of competing interest

None declared.

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